OPEN AGENT HACKATHON 2026 · TRACK 1 — SOLVING FRAGMENTED INTELLIGENCE

# RecallWhy

Agents that tell a retailer which recalled products are on its shelves, which lots, which stores, how many customers to call — **and why they believe it.**

Zetaris federated SQL · Meterless H-MEM + World Model · Meterless Nightshift · NVIDIA Nemotron 3 Ultra · built in Cursor

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github.com/mikey92/recallwhy

# 517 federal recall notices since July. Each one is a manual cross-check.

A regional retailer's **product-safety / QA manager** reads every CPSC and FDA notice and must answer:

- Do we sell this — or something that only looks like it?
- Which lots? Which stores still hold them?
- How many customers bought it since the recall began?
- Who supplied it?

**184 + 333**

CPSC recalls and FDA food enforcement reports, 1 Jul – 21 Oct 2026, read live by RecallWhy

The answers live in four systems that were never meant to meet — two **recall feeds**, the **ERP** and the **supplier portal** — each writing a product its own way: UPC-12 in the ERP, GTIN-14 from a third of suppliers, digits buried in notice text. A wrong "not ours" leaves a contaminated lot on the shelf; a wrong "ours" pulls good stock.

2 · SOLUTION

# Five agents map the fragments, then investigate every recall and return a verdict with its evidence.

### Cartographer

Lists every Zetaris source and column, profiles identifiers, tests cross-source joins. Finds the GTIN-14 suppliers upfront. When the Auditor hits rows the key can't reach, it diagnoses that supplier, tests the fix on all its ship notices, and corrects the map.

### Scout

Reads both live feeds and the item master through Zetaris; one federated UPC sweep; opens a case per recall that touches the catalog.

### Investigator

UPC / model rules, Nemotron judge for text matches, lot codes, prior decisions from H-MEM, exposure by store and lot with federated SQL.

### Auditor

**Can block.** Re-checks the live notice, quotes and lots verbatim, re-derives every number with its own SQL, checks each brief sentence. ≤2 revisions, then a human.

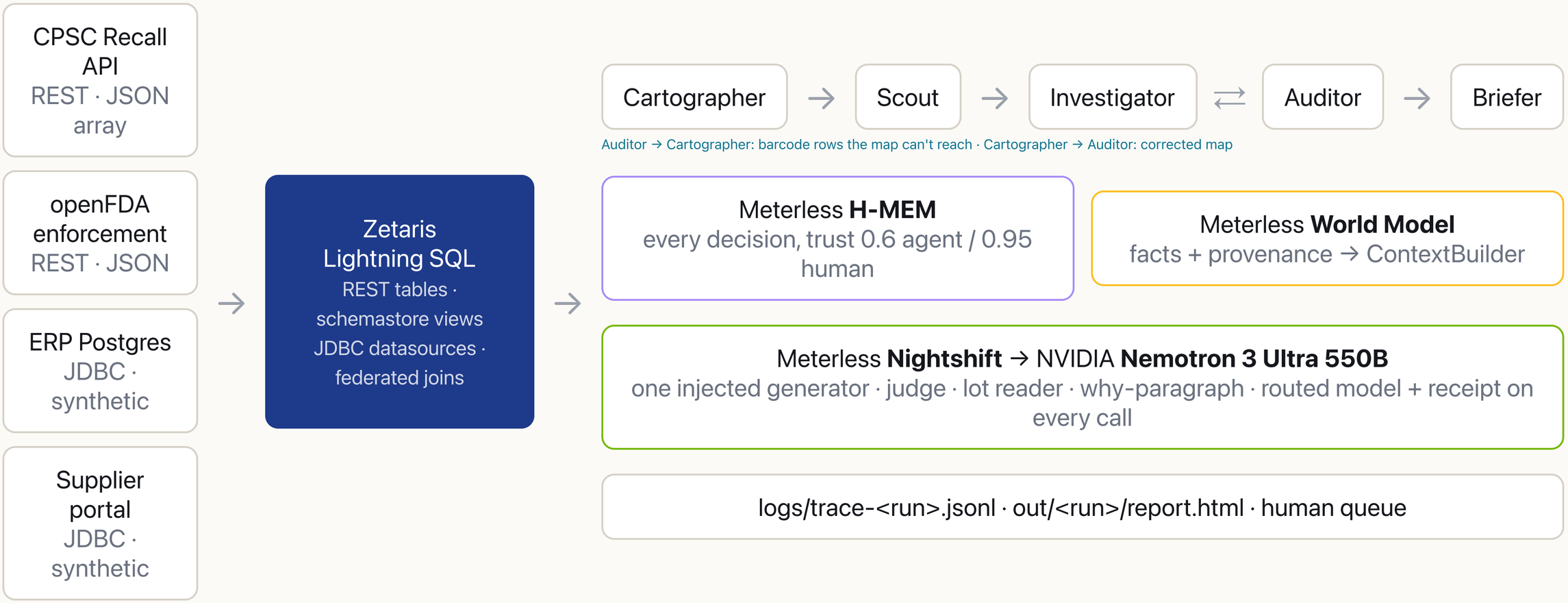
### Briefer

Writes only from the World Model context: numbered evidence, each with the SQL or model call behind it.

ACT NOW | NOTIFY | NO EXPOSURE | NOT OURS | NEEDS HUMAN

3 · ARCHITECTURE

# Every agent reads data only through Zetaris.



4 · STACK

# Small, typed, runnable from a clean clone.

| LAYER | WHAT |
| --- | --- |
| Data access | Zetaris Lightning SQL over its REST API ( `/api/proxy/sql-editor/sqls/run-query` ) |
| Agents | TypeScript on Node 22, no agent framework — five agents, one orchestrator, one trace |
| Memory & state | Meterless H-MEM and World Model reference engines (Apache-2.0), mounted in `src/engines/` |
| Model | Meterless **Nightshift** Runtime API (OpenAI-compatible) → NVIDIA **Nemotron 3 Ultra 550B-A55B**; no model and a human queue when Nightshift is absent |
| Retailer data | Postgres 16 in Docker — synthetic ERP + supplier portal (seeded generator) |
| Entry points | `POST /run` API · Dockerfile · CLI · live trace viewer |
| No-credential mode | `SAMPLE_MODE=true` : same agents and SQL on an in-process DuckDB stand-in, labelled SAMPLE |
| Development | Cursor (agent scaffolding, generator, SQL layer, tests) · 44 tests incl. Meterless engine tests |

5 · HOW ZETARIS AND METERLESS ARE USED

# Load-bearing, not bolted on.

## Zetaris

- Cartographer maps sources with Zetaris's catalog: `SHOW DATASOURCES`, `DESCRIBE DATASOURCE TABLE`, `SHOW LIGHTNING DATABASES` → profiles and join tests
- 2 live Lightning REST tables → schemastore views
- 2 JDBC datasources (ERP, supplier portal)
- Scout: **REST × JDBC** join — live FDA notice digits ⋈ ERP UPCs
- Investigator: ship notices ⋈ suppliers ⋈ ERP item master
- Auditor: re-queries the live notice and re-aggregates every number itself
- Setup is code: `npm run setup:zetaris`

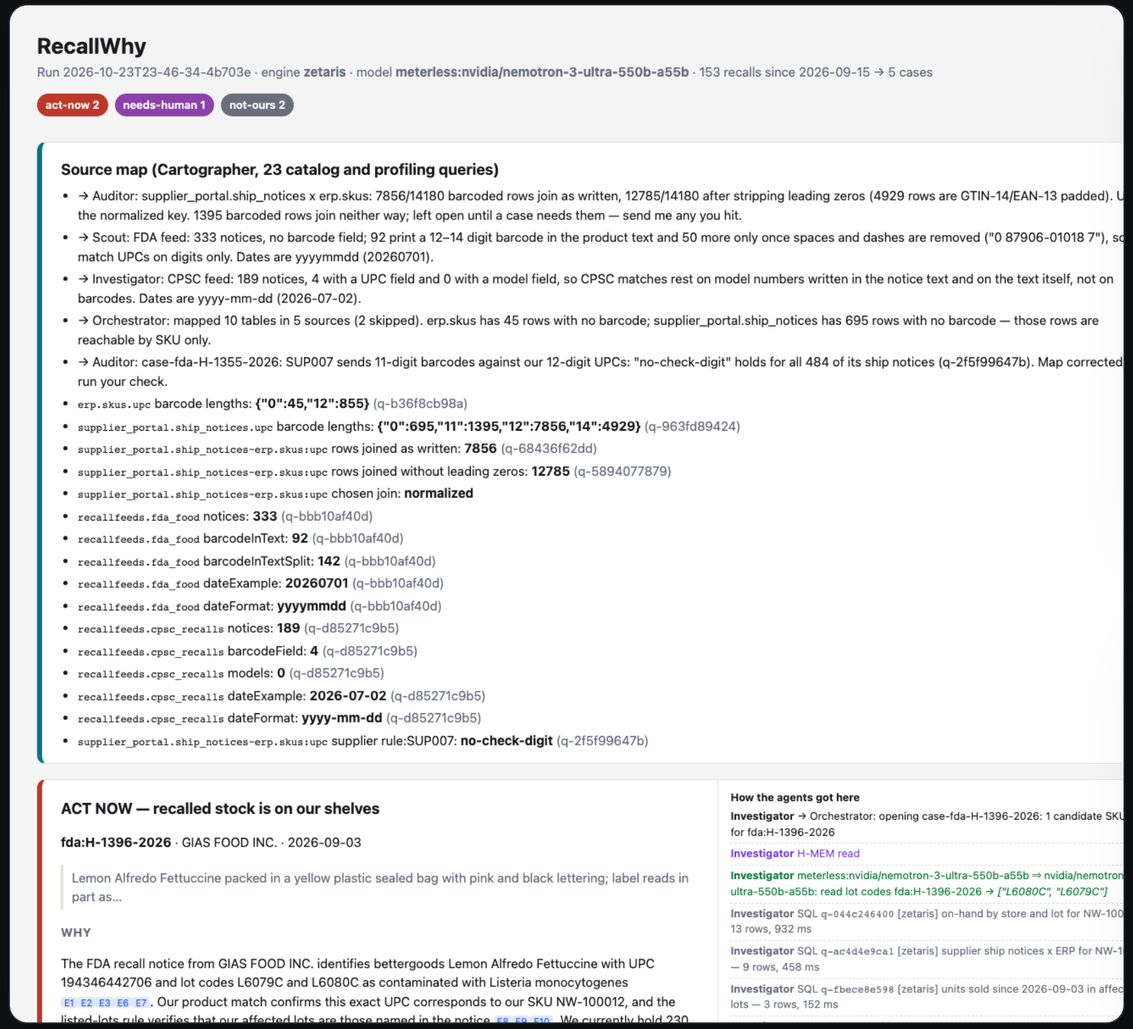
## Meterless

- **H-MEM**: every adjudication stored with trust; human decisions (0.95) are followed on the next run; audited agent decisions are reused without a model call; trust ledger logs reads/writes
- **World Model**: recalls, cases, findings as facts; every write carries provenance (agent, SQL id, model call)
- **ContextBuilder** is the Briefer's only input — it cannot cite what the team didn't establish
- **Nightshift** is the primary AI provider: every model call, with its receipt and routed model in the trace

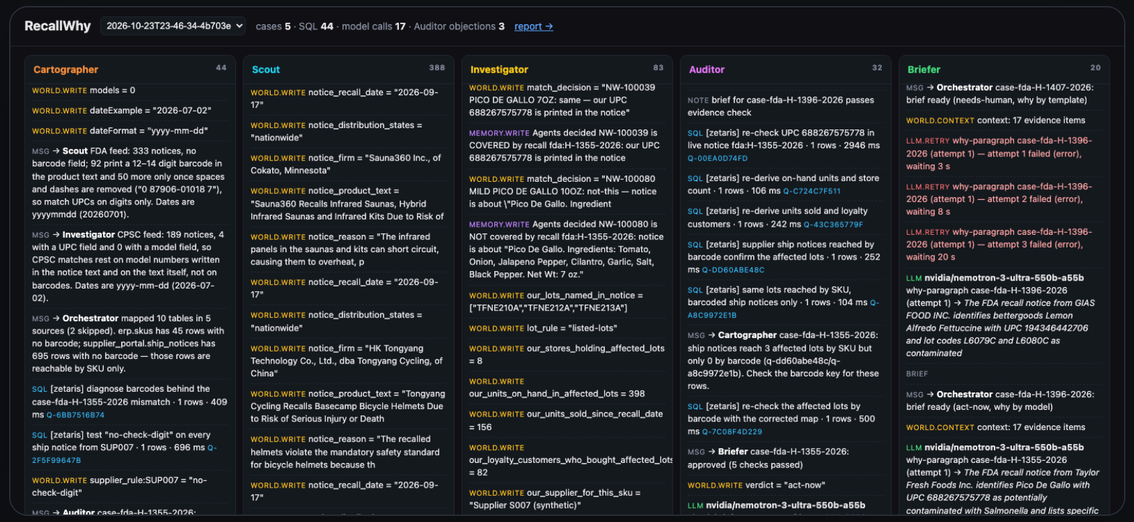
# Map the fragments first. Then trust no one — including the model.

- **The map is a hypothesis the Auditor can send back.** Rows the key can't reach go to the Cartographer, which learns the supplier's rule (padding, dropped check digit) and proves it on all that supplier's ship notices. Without the Cartographer, rules-only matching catches **12** of 194 covered pairs instead of **103**.
- **Re-derives every number in Zetaris** with differently written SQL; a mismatch blocks the brief.
- **Re-reads the live notice** (REST table) to confirm a UPC match, instead of trusting the parser.
- **"Cheese ≠ pepperoni"**: a quote can be copied from the notice and still describe another variety — every product word of our item must appear in the notice. Live run: blocked, re-judged, ruled *not ours*.
- **Lot codes the model reads must appear verbatim** in `code_info`; invented lots are dropped.
- **Brief sentences are checked** against the evidence they cite — it caught a draft that invented stores in Kentucky and Ohio.
- **Human decisions change the next run** through H-MEM, and the brief cites them.

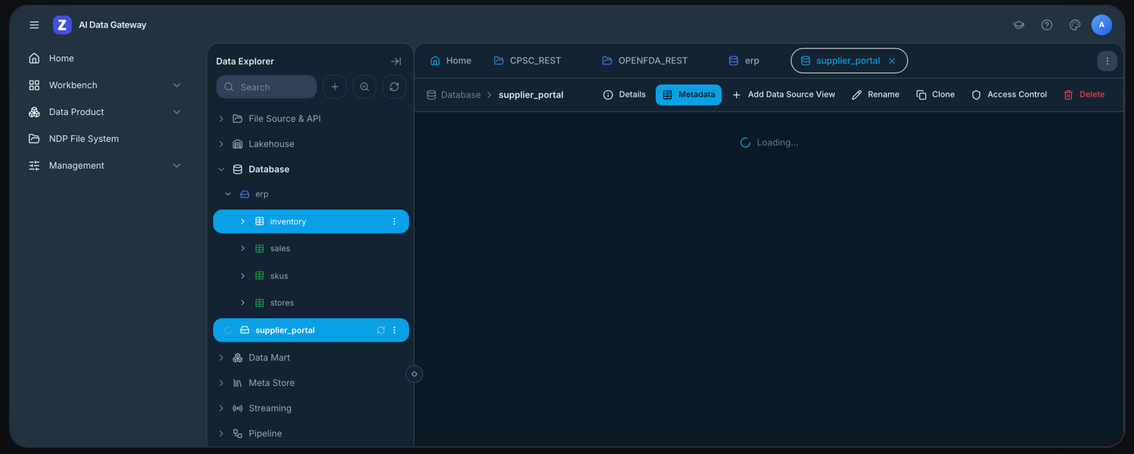
# 7 · PRODUCT




Brief: verdict, why (cited), actions, caveats, evidence with SQL ids — and the agent trace that produced it.




Live viewer: the agents' events as they happen, from the trace file.




Zetaris: REST feeds and JDBC sources registered by `setup:zetaris`.

8 · RESULTS

# 186 of 194 recalled items caught. The Auditor cuts missed recalls 12 → 4.

| SETUP (440 RECALL/SKU PAIRS) | CAUGHT (OF 194) | WRONGLY DISMISSED | FALSE ALARMS (OF 246) | TO HUMAN |
|---|---|---|---|---|
| No Cartographer — rules only | 12 | 0 | 0 | 428 |
| Cartographer — rules only | 103 | 0 | 0 | 337 |
| + Nemotron 3.5 Lightning 30B via Nightshift | 141 | 10 | 2 | 63 |
| **+ NVIDIA Nemotron 3 Ultra via Nightshift** | **186** | **4** | **6** | **23** |

| TEXT MATCHES (90 COVERED OF 336) | CAUGHT | WRONGLY DISMISSED | FALSE ALARMS | TO HUMAN |
|---|---|---|---|---|
| Ultra: judge's first answer | 78 | 12 | 9 | 0 |
| **Ultra: after the Auditor loop** | **84** | **4** | **6** | **21** |

A missed recall leaves stock on the shelf, so the Auditor checks dismissals too: a dismissal whose own quote names our product is sent back, then to a human. 0 of 537 model calls failed (rate limits retried).

Real Investigator ⇄ Auditor loop on every candidate pair from 517 recalls (sample-mode data). Covered = SKU generated from that recall, or its UPC printed in the notice. Retailer data is synthetic. Details: eval/EVAL.md.

9 · NEXT

# From demo to a safety desk.

- More feeds through Zetaris: USDA FSIS, NHTSA equipment (car seats), Health Canada
- Lot capture from receiving scans instead of the item master
- Customer notification through the loyalty system — counts today, messages tomorrow
- Cartographer onboarding new feeds on its own: catalog → profile → join test
- Markovian engine for daily long-running sweeps
- Review queue UI with one-click decide → H-MEM

**Limits today:** synthetic retailer data; lexical Auditor checks; CPSC rarely prints UPCs; hosted model rate limits make a full evaluation take two hours.